Intro to ECO 3253

Economics of Public and Social Issues

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ECO3253, UTSA

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What is this course about?

This course

Welcome to **ECO 3253**!

Big picture:

- How to use data to understand and solve current and important economic problems!
- Get a sense of research frontier in applied economics and social science (equality of opportunity and mobility, education, innovation and entrepreneurship, health care, climate change, and crime, etc)
- How will we get there? By doing 3 things:
  1. covering the topics with a focus towards using data that can help answer these questions
  2. understanding the intuition for how to use data to answer these questions (basics of statistical analysis)
  3. using computational tools to help us with the statistical analysis (basics of R)

About me

Let me briefly tell you about me:

- Colombian
- PhD in Economics at Duke University in North Carolina
- Masters in Economics at Université Catholique de Louvain in Belgium
- Undergraduate in (you guessed it!) Economics at Universidad Nacional de Colombia.
- Applied (micro)economist:
  - 'micro': focus on specific markets (housing and media)
  - 'applied': I use data all the time

Website

We will have a cool website with the materials covered in class:

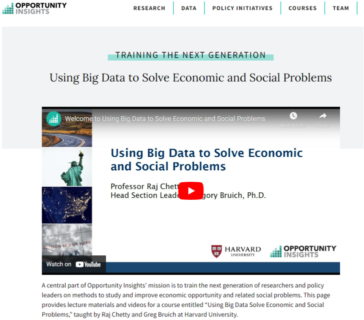
https://jrm87.github.io/ECO3253_spring2026/

Info on Canvas. Write it down. Keep it in mind!

Topics to be covered - 1

Lectures based on the course by Prof. Raj Chetty.

R: Chetty	Opportunity Atlas Course



Topics to be covered - 2

The plan for what we will covered in the semester includes:

- Geography of Upward Mobility in America
- Causal Effects of Neighborhoods and Characteristics of High-Mobility Areas
- Historical and International Evidence on the Drivers of Inequality and Mobility
- Upward Mobility, Innovation, and Growth
- Higher Education and Upward Mobility
- Primary education
- Teachers and Charter Schools
- Racial Disparities in Economic Opportunity
- Improving judicial decisions
- Immigration
- Political Economy
- Income taxation
- Savings and wealth
- Housing markets and COVID
- Intro to air and water pollution, and externalities

Data-driven course

- Data driven, not just theoretical
- We will work with real world data, and we will try to make sense of it.
- Economics is much more empirical now than a few years ago

Number of data-driven articles in leading journals

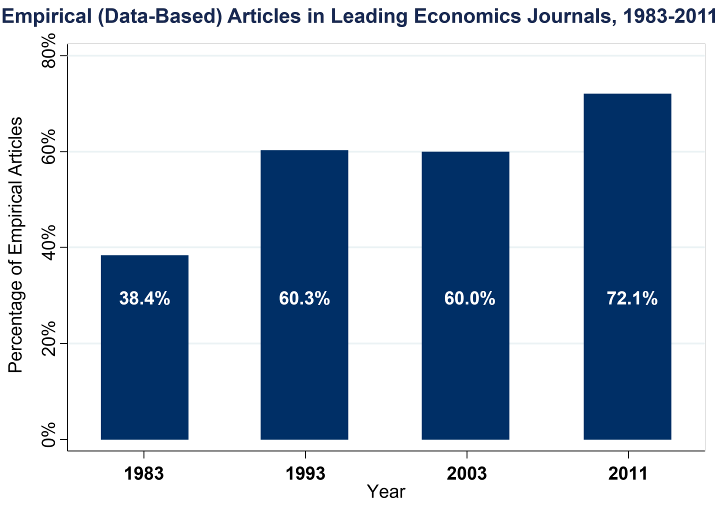


Figure 1: Source: Mamermesh (JEL 2013)

What do we need to understand data?

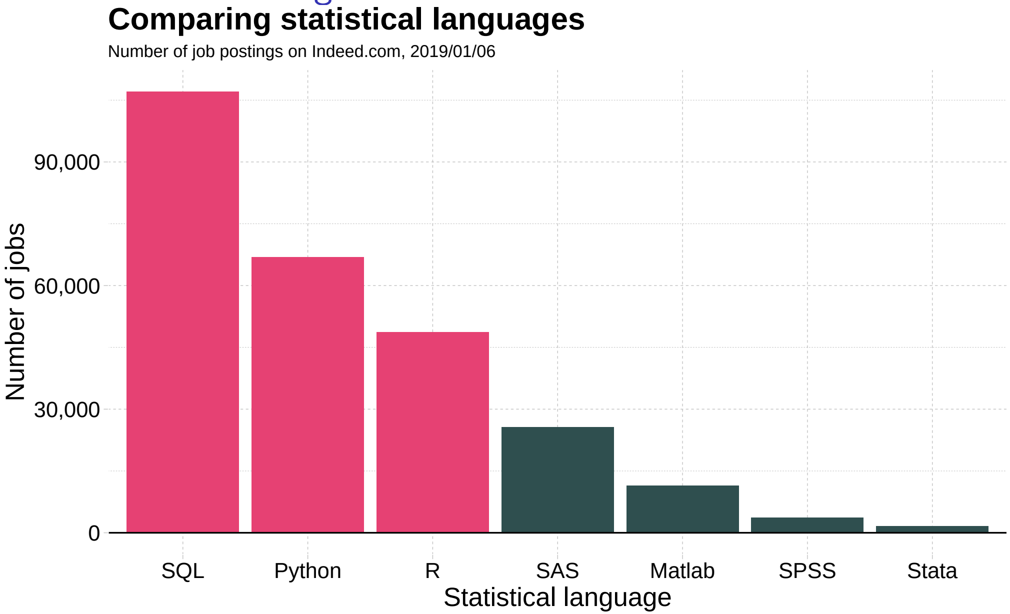
How can we know if going to school increases wages? Or if economic mobility is low or high?

- Framework (basic stats and intuition)
  - I will provide all the basic intuition for the empirical methods. These include:
    - Descriptive Data Analysis: correlation, regression
    - Experiments: randomization, non-compliance
    - Quasi-Experiments: regression discontinuity, difference-in-differences
    - Machine Learning: prediction, overfitting, cross-validation
- A powerful (and very cool) calculator:
  - **R** as our statistical software
  - Will follow parts of the book ModernDive
  - I will guide you through it - does not matter if you have no experience with 'programming'

Ok, but why R?

- **R** is free and open source!
- **R** has a vibrant online community!
- **R** is very flexible and powerful — adaptable to nearly any task (*e.g.*, correlations, econometrics, spatial data analysis, machine learning, web scraping, data cleaning, website building, teaching.)
- Employers like **R** over alternatives

Added benefits of learning R



R, RStudio and Posit Cloud

- We will talk more about R and RStudio next Monday.
- I will reach out with very simple instructions sometime this week.
- You will not have to install **anything** in your computer.
  - Will use cloud services. More on that later.

Economic Concepts

Back to the content! We will cover and use several economic concepts you probably learned before. We will see them in practice:

1. Effects of price incentives
2. Supply and demand
3. Competitive equilibrium
4. Adverse selection
5. Behavioral economics vs. rational models

Grades

Activity	Perc. Weight
Projects 0–4	20%
Midterm	25%
Final	25%
Attendance and participation	30%
Total	**100%**

Projects

- A big part of the course: **4 projects** (plus a warm-up "project 0")
- More involved than typical homeworks: real data, real problems, real code
- Submitted on **Canvas** by **11:59 pm** on the due date
- **How they are graded:**
  - Each on-time submission earns **full credit** (✓) toward the **Projects 0–4 (20%)** bucket — I do not line-by-line correct them
  - The **substance** of what you submitted (your reasoning, assumptions, code choices, interpretation) is evaluated separately, through the **Attendance & participation (30%)** bucket — see next slide
- You may use generative AI tools, but you are responsible for verifying outputs and being able to explain every line you submit
- Start early! Many new elements — give yourself time to try, fail, and experiment
  Do not work on these projects the day before!

Participation: how the 30% works

This is a face-to-face course built around active discussion and frequent reference to the projects you submit.

- On any class day, I may **randomly call on you** to:
  - answer or expand on the topic we are covering in lecture, or
  - **explain a piece of your submitted project** (code logic, identification assumptions, interpretation of a result)
- If you prefer, you can "**call on a friend**" — a classmate volunteers to help expand the answer
- If you are **called and absent** without documentation, that missed opportunity counts toward the participation score (see UTSA policies for excused absences)

What strong participation looks like

- **Preparation & relevance.** You come prepared, reference the assigned readings or methods, and tie comments back to your own submitted work
- **Reasoning & evidence.** You state your assumptions, defend your identification strategy, and recognize when an estimate or AI suggestion needs to be verified
- **Code fluency.** You can walk through key parts of your code (data wrangling, estimation, diagnostics), make a small on-the-spot edit, and explain *why* it addresses the question
- **Dialogue.** Respectful engagement, clarifying questions, willingness to call on a friend to extend a point